

Relaxing the Constraints: Working from Home and Labour Supply Preferences^{*}

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Abstract

Working from home (WFH) has expanded persistently since the COVID-19 pandemic, yet whether it relaxes labour-supply constraints or instead intensifies work remains unclear. We study this question using the 2023 wave of the German Socio-Economic Panel (SOEP), which uniquely records both realised and desired working hours in the post-pandemic period, and instrumenting realised WFH with pre-pandemic occupational WFH feasibility. We find that WFH increases both realised and desired labour supply for women, while for men it raises realised hours but leaves desired hours unaffected. The gender divergence points to distinct mechanisms. For women, the increase in desired hours is consistent with WFH relaxing time-coordination constraints between market work and home production. For men, the rise in realised hours without a corresponding increase in desired hours is more consistent with work intensification. Our findings suggest that remote work converts previously unrealised labour supply preferences into realised hours among workers constrained by household responsibilities, with important implications for gender equality and labour shortages in ageing economies.

Keywords: working from home, labour supply, gender gaps, SOEP

JEL codes: J22, J16, J81, D13, C26

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1. Introduction

The COVID-19 pandemic triggered a rapid and large-scale expansion of working from home (WFH), transforming where and how a substantial share of the labour force performs its tasks. This shift has proved remarkably persistent. Survey evidence from more than 25 countries indicates that, by 2023, employees worked from home on average roughly 1.5 days per week, far above the pre-pandemic norm and with little sign of reversal (Aksoy et al. 2025; Barrero, Bloom, and Davis 2021). Germany stands out as one of the continental European countries with the highest rates of WFH adoption (Aksoy et al. 2023), with approximately one in three employees working from home at least once per week in 2023.

A growing literature studies the consequences of this transformation for labour market outcomes and generally finds that WFH increases labour supply. However, the mechanisms underlying this effect remain unclear. On the one hand, WFH may relax time-allocation constraints by increasing scheduling autonomy and facilitating the coordination of market work and home production. Under this *flexibility channel*, workers may be able to supply hours that were previously infeasible. On the other hand, WFH may intensify work by blurring the boundaries between work and private life, reducing breaks, and increasing expectations of availability. Under this *work intensification channel*, workers may end up working longer hours without necessarily preferring to do so. Distinguishing between these mechanisms is important both for understanding the welfare implications of remote work and for assessing its potential role in shaping labour supply.

We distinguish between these mechanisms using desired working hours. Standard labour supply models assume that workers freely choose hours to equate the marginal utility of leisure to the real wage. In practice, however, hours are often constrained by employers, schedules, and coordination frictions (Kahn and Lang 1991; Labanca and Pozzoli 2022), generating persistent gaps between usual and desired hours worked (Beckmannshagen and Schröder 2022; Jarosch, Pilossoph, and Swaminathan 2025). If WFH relaxes such constraints, workers should not only work more, but also report wanting to work more. By contrast, if WFH primarily intensifies work, usual hours may rise while desired hours remain unchanged. We therefore study whether WFH affects both realised and desired labour supply.

The flexibility channel is likely to matter especially for workers facing strong time-coordination frictions between market work and home production. A large literature documents that women, and mothers in particular, bear a disproportionate share of child-care and household responsibilities, which constrains labour supply and contributes to gender inequalities in the labour market (Goldin 2014; Kleven, Landaïs, and Sogaard 2019). The traditional workplace model, characterised by fixed schedules and physical presence requirements, is often difficult to reconcile with these responsibilities. Consistent with this view, previous research shows that women place particularly high value on flexible

work arrangements (Mas and Pallais 2017), while rigid work schedules and non-linear returns to long hours are central drivers of gender gaps in earnings and career progression (Goldin 2014; Cortés and Pan 2019). In this context, WFH may disproportionately expand the feasible set of labour supply choices for time-constrained workers. This motivates our central research question, that is whether WFH increases desired labour supply, and whether this effect differs by gender.

To answer this question, we use data from the 2023 wave of the German Socio-Economic Panel (SOEP). In addition to detailed socio-demographic and labour-market information, the SOEP contains a measure of desired weekly working hours. Specifically, respondents are asked how many hours they would choose to work if their income adjusted proportionally to hours worked.¹ We interpret this variable as capturing optimal labour supply conditional on wages and time constraints. Importantly, the SOEP is, to the best of our knowledge, the only major household panel that simultaneously records both desired working hours and detailed WFH arrangements in the post-pandemic period. We consider both the extensive margin of WFH (whether the individual works from home at least once per week) and the intensive margin (weekly hours worked from home).

A central empirical challenge is that WFH adoption is endogenous. Workers are not randomly assigned to remote work arrangements: individuals with stronger preferences for flexibility, tighter household constraints, or higher unobserved productivity may be both more likely to select into WFH and more likely to adjust their labour supply. As a result, ordinary least squares (OLS) estimates may be biased in either direction. For instance, if time-constrained individuals disproportionately select into WFH, OLS will understate its true effect on hours; conversely, if more productive or career-oriented workers sort into WFH jobs, OLS may overstate it. To address this issue, we instrument realised WFH using a pre-pandemic measure of occupational WFH feasibility developed by Alipour, Falck, and Schüller (2023) from the 2018 BIBB/BAuA Employment Survey, following Alipour (2025) and Meister, Menkhoff, and Schröder (2025). The instrument captures the technological feasibility of performing job tasks remotely at the occupational level prior to the pandemic-induced expansion of WFH. Conditional on controls and federal-state fixed effects, the identifying assumption is that pre-pandemic WFH feasibility affects labour supply outcomes only through realised WFH adoption. This is plausible for two reasons. First, the measure is predetermined and reflects task content rather than individual choices, mitigating concerns about reverse causality. Second, conditional on occupation and worker characteristics, it is unlikely to be directly correlated with unobserved time-varying shocks to individual labour supply preferences in 2023. At the same time, the instrument is strongly relevant as occupations with higher pre-pandemic feasibility experienced substantially larger increases in WFH during and after

¹See Section 2.1 for further details on the “desired hours” variable.

the pandemic.

Our results reveal a sharp gender asymmetry. For women, WFH substantially increases both usual and desired weekly hours worked. The estimated effects on desired hours are large and close in magnitude to those on usual hours, suggesting that WFH not only changes where work is performed, but also relaxes constraints limiting desired labour supply. For men, by contrast, WFH increases usual hours while leaving desired hours largely unaffected, consistent with a reorganisation or intensification of work rather than an expansion of preferred labour supply.

Our paper contributes to the growing literature on remote work and labour supply in two ways. First, we provide the first evidence that WFH raises desired labour supply for women, and not for men. By jointly analysing usual and desired hours, we provide direct evidence consistent with the flexibility interpretation of remote work for time-constrained workers. Second, our findings speak to an important dimension of gender inequality in labour market participation, as the results suggest that many women would prefer to supply more labour, but face constraints that standard workplace arrangements do not accommodate. To the extent that WFH relaxes these constraints, remote work may help mobilise labour supply among groups whose participation is limited by time-allocation frictions, an issue of increasing relevance in ageing economies facing labour shortages.

Related Literature. This paper relates to two strands of literature: the growing body of work on the labour-market consequences of remote work, and the established literature on gender gaps in labour supply and the role of workplace flexibility.

Early causal evidence on WFH comes from [Bloom et al. \(2015\)](#), who conduct a randomised controlled trial at a Chinese call centre and find that WFH raised productivity by 13% and halved attrition, while reducing promotion rates conditional on performance. Using pre-pandemic SOEP data, [Arntz, Yahmed, and Berlingieri \(2022\)](#) document that WFH is associated with longer working hours, especially for mothers, but that these additional hours are not rewarded with higher wages for women. The pandemic-induced expansion of remote work has generated a new body of evidence on a broader set of outcomes, including productivity, hours worked, and job mobility ([Gibbs, Mengel, and Siemroth 2023](#); [Angelici and Profeta 2024](#); [Bloom, Han, and Liang 2024](#)), as well as the motherhood penalty in employment ([Emanuel and Harrington 2024](#)), home production, and intra-household time allocation ([Alipour 2025](#); [Feuillade, Goux, and Maurin 2025](#); [Schüller 2026](#)). [Alipour \(2025\)](#), using SOEP data and a similar identification strategy to ours, finds that WFH reinforces gender gaps in unpaid work, as women who work from home substantially increase domestic labour and caregiving, while men's time-allocation response is small, and gender gaps in both paid and unpaid work widen when partners jointly adopt WFH. [Feuillade, Goux, and Maurin \(2025\)](#) study spousal cross-effects using

French administrative data on collective WFH agreements, finding that one partner’s shift to remote work raises the other partner’s WFH intensity, with asymmetric effects that depend on relative earnings and the domestic division of labour. [Schüller \(2026\)](#) estimates the effect of fathers’ WFH on the parental division of housework and childcare in Germany, documenting a 21 percentage-point reduction in the maternal share of housework, while childcare allocation remains persistently traditional.

These gender asymmetries are tightly connected to a well-established literature linking gender gaps in labour market outcomes to the structure of working-time demands. [Goldin \(2014\)](#) argues that non-linearities in the returns to hours worked are a primary driver of the residual gender wage gap. As employers reward long and inflexible hours disproportionately, occupational structures that penalise schedule flexibility generate systematic earnings penalties that fall disproportionately on women. [Cortés and Pan \(2019\)](#) extend this analysis to occupational choice, showing that the prevalence of non-linear pay schedules shapes sorting across jobs and contributes to the gender composition of high-earnings occupations. On the preference side, [Mas and Pallais \(2017\)](#) finds that women are willing to accept substantially lower wages in exchange for scheduling flexibility and the option to work from home. [Chen et al. \(2023\)](#) confirm and extend this finding, documenting that the premium placed on WFH is especially large for women with children.

Our paper bridges these two strands by providing the first evidence that WFH raises desired, and not only realised, labour supply, with effects concentrated among women. To our knowledge, desired hours have not previously been studied in the WFH literature, and their availability in the SOEP makes Germany a uniquely suitable setting for this analysis. By comparing movements in desired and usual hours, we can moreover distinguish empirically between the flexibility channel and the work-intensification channel of remote work, so to understand whether WFH expands workers’ feasible set or merely reorganises how existing preferences are met.

The remainder of the paper is organised as follows. Section 2 describes the data, sample construction, and descriptive statistics. Section 3 presents the empirical strategy and discusses the identification assumptions. Section 4 reports the main results, including gender heterogeneity, while Section 5 presents robustness analyses. Section 6 concludes.

2. Data, Sample Construction, and Descriptive Analysis

2.1. SOEP

The main data source for our analysis is the German Socio-Economic Panel (v40.1), a nationally representative annual household panel for Germany. In 2023, the survey inter-

viewed approximately 10,000 households.² The richness of the SOEP allows us to observe a wide range of socio-economic characteristics, detailed household information, and labour market histories. Crucially for our purposes, the survey provides measures of both usual and desired weekly working hours, alongside detailed information on WFH arrangements, including both WFH participation and weekly hours worked from home. The availability of desired working hours is particularly important for our analysis as, to the best of our knowledge, the SOEP is the only major household panel that records desired hours after COVID-19.³ Moreover, Germany provides a particularly relevant setting as it combines high WFH adoption with pronounced gender gaps in employment, also favoured by strong institutional features like the joint income taxation system (*Ehegattensplitting*), that have historically reinforced within-household labour specialisation, making the time-coordination constraints relaxed by WFH especially salient.

Measures of usual and desired hours of work have been collected consistently since the first SOEP wave in 1984. Both variables refer to weekly hours. Usual hours are defined as contractual working hours plus any paid or unpaid overtime usually worked. Desired hours are elicited through the following question:

“If you could choose your own working hours, taking into account that your income would change according to the number of hours: How many hours would you want to work?”

This formulation explicitly links hours adjustments to proportional income changes, discouraging respondents from reporting lower preferred hours under the assumption that earnings would remain unchanged. In addition, [Beckmannshagen and Schröder \(2022\)](#) show that stated desired hours strongly predict subsequent labour-market behaviour, suggesting that these measures contain economically meaningful information about workers’ labour supply preferences.

Turning to the WFH variables, in 2023 respondents were asked how often they had worked from home in their current job during the previous six months. Following [Alipour \(2025\)](#), we construct a dummy equal to one for individuals who report working from home at least once per week. This variable captures WFH at the extensive margin. To measure the intensive margin, we use the number of hours worked from home per week on average, elicited through a separate question in the same survey wave.

Appendix Section A.1 provides more information on the relevant variables.

²For an overview of the SOEP, see, for example, [Schröder et al. 2020](#).

³For example, the British Household Panel Survey (BHPS) collected desired hours only up to 2008, while the US Current Population Survey (CPS) included comparable questions only in selected supplements in 1985 and 2001.

2.2. WFH feasibility index

As a measure of our instrument, WFH feasibility, we use the index of work-from-home capacity for the German economy developed by [Alipour, Falck, and Schüller \(2023\)](#). The index is constructed using the 2018 wave of the BIBB/BAuA Employment Survey, a nationally representative survey of approximately 17,000 employees in Germany that collects detailed information on the task content of jobs. The key building block is a self-reported feasibility indicator at the individual level, as a job is classified as WFH feasible if the respondent does not rule out the possibility of working from home, either because she currently works from home or because she would accept a temporary offer to do so. The individual-level indicator is then aggregated to the occupational level using employment weights, yielding the share of workers in each occupation whose job can at least partially be performed from home. In our main analysis, we employ the index aggregated at the 2-digit ISCO-08 level. This instrument has been adopted in related work studying the consequences of remote work on gender inequality in paid and unpaid labor ([Alipour 2025](#)) and on stock market participation ([Meister, Menkhoff, and Schröder 2025](#)).

2.3. Sample

The year of interest for our analysis is 2023, which we treat as a cross-section. Our identification strategy, however, relies on pre-pandemic WFH feasibility at the occupational level. To mitigate concerns that our estimates are driven by workers sorting into occupations with higher (or lower) post-pandemic WFH feasibility, we restrict the sample to individuals who are observed in every year from 2019 to 2023 and who do not change jobs over this period. We further restrict the sample to individuals who, in 2023, are aged between 18 and 65, are employees, and have non-missing values for our key outcome and identification variables: usual hours of work, desired hours of work, the WFH dummy, hours worked from home, the WFH feasibility index, and occupational category.⁴ Finally, we exclude a small number of implausible observations, such as full-time workers reporting fewer than 30 weekly hours or zero desired hours of work. These restrictions yield a final sample of 2,627 individuals (1,254 men and 1,373 women). Appendix Table A.1 reports the evolution of the sample size at each stage of the sample selection.

Table 1 reports summary statistics for the estimation sample separately for men and women, as our main results are also presented by gender. Men and women are similar in observable characteristics: both groups are on average around 50 years old, about one sixth have a migration background, and educational attainment is comparable, with roughly 13.5 years of schooling on average. Men are, however, somewhat more likely to hold a university degree (40% vs. 34%). They are also more likely to be married (62% vs. 54%) and

⁴The WFH feasibility index is not available for military occupations, which in any case represent a very small share of the sample.

to live with an underage child (42% vs. 36%), consistent with the sample being restricted to employed individuals, hence excluding housewives. In contrast, labour market outcomes differ markedly by gender. Part-time employment is substantially more prevalent among women (53%) than men (9%). Consistent with these differences, men earn on average about 50% more in monthly net earnings and receive higher hourly wages. Men also report longer working hours, both in terms of usual and desired hours, indicating a sizable gender gap in labour supply. Turning to working from home, 33% of men and 29% of women report working from home at least once per week in 2023. Conditional on working from home, men also report slightly higher weekly hours worked from home. Pre-pandemic WFH feasibility is also nearly identical across genders, suggesting no systematic gender differences in sorting into more or less WFH-feasible occupations.

2.4. Some facts on WFH and working hours

Our empirical analysis starts with a set of descriptive facts about the relationship between WFH and working hours across household types. If WFH alleviates the burden of home production and childcare by improving time coordination, the groups that appear to benefit the most in terms of labor supply should also be those making more intensive use of it.

Figure 1 reports the average WFH intensity in 2023, defined as the share of weekly hours performed from home, across four demographic groups capturing different degrees of household time constraints (overall population, single individuals, cohabiting couples, and cohabiting couples with children), separately by gender.⁵ A clear pattern emerges for women: WFH intensity rises monotonically with household and childcare responsibilities, from 14.8% among single women to 15.2% for married women, and reaching 18.7% among married women with children. No comparable pattern holds for men, whose higher WFH intensity in cohabiting couples likely reflects occupational sorting towards non-routine cognitive jobs which are more easily performed remotely, as married men are traditionally more specialized in market work and career progression. At this point, it is worth examining more closely the relationship between WFH and working hours, taking into account this third dimension of household responsibilities.

Figure 2 reports mean usual weekly hours by demographic group in 2023, distinguishing teleworkers from non-teleworkers.⁶ Within each group, circles denote the group of

⁵In detail, we measure WFH intensity within group g as the ratio between average weekly hours worked from home and average total weekly working hours:

$$\text{WFH Intensity}_g = \frac{\overline{\text{WFH hours}_g}}{\overline{\text{weekly hours}_g}} \times 100.$$

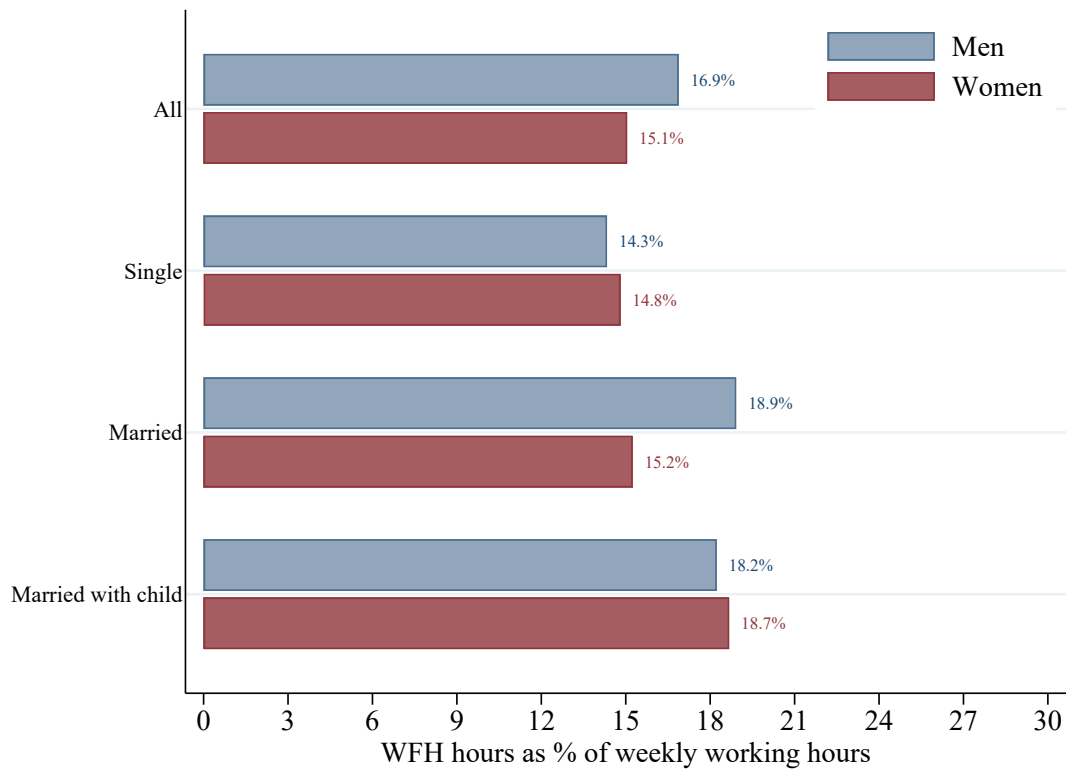
⁶In particular, we identify teleworkers through the SOEP variable *plb0732v2*, which indicates the frequency of working from home over the past six months. An individual is classified as a WFH individual if she has

TABLE 1. Summary statistics

	Men	Women
<i>Panel A. Labour market outcomes</i>		
Part-time employed	0.09 (0.29)	0.53 (0.50)
Net monthly earnings	2756.95 (1324.62)	1830.71 (955.03)
Net hourly wage	15.28 (6.61)	12.53 (5.36)
Actual weekly hours	41.45 (7.80)	33.84 (10.62)
Desired weekly hours	34.35 (9.86)	28.44 (10.04)
WFH at least once per week	0.33 (0.47)	0.29 (0.45)
WFH hours per week	7.66 (12.71)	5.60 (10.69)
WFH feasibility (2019)	65.96 (28.40)	66.29 (24.75)
Work experience (years)	24.98 (10.98)	23.87 (9.99)
<i>Panel B. Demographics</i>		
Age	48.92 (10.95)	50.19 (9.85)
Migration background	0.17 (0.38)	0.16 (0.37)
Married	0.62 (0.49)	0.54 (0.50)
Child under 18 in HH	0.42 (0.49)	0.36 (0.48)
Years of education	13.54 (2.81)	13.45 (2.62)
University degree	0.40 (0.49)	0.34 (0.47)
Observations	1254	1373

Notes: This table reports averages and standard deviations in parentheses by gender for the estimation sample in 2023. Summary statistics are not weighted. The sample is restricted to individuals aged 18-65 who are employed and have not changed jobs since 2019. Earnings are expressed in 2020 prices. “HH” stands for household. Data source: SOEP 2023 (v40.1).

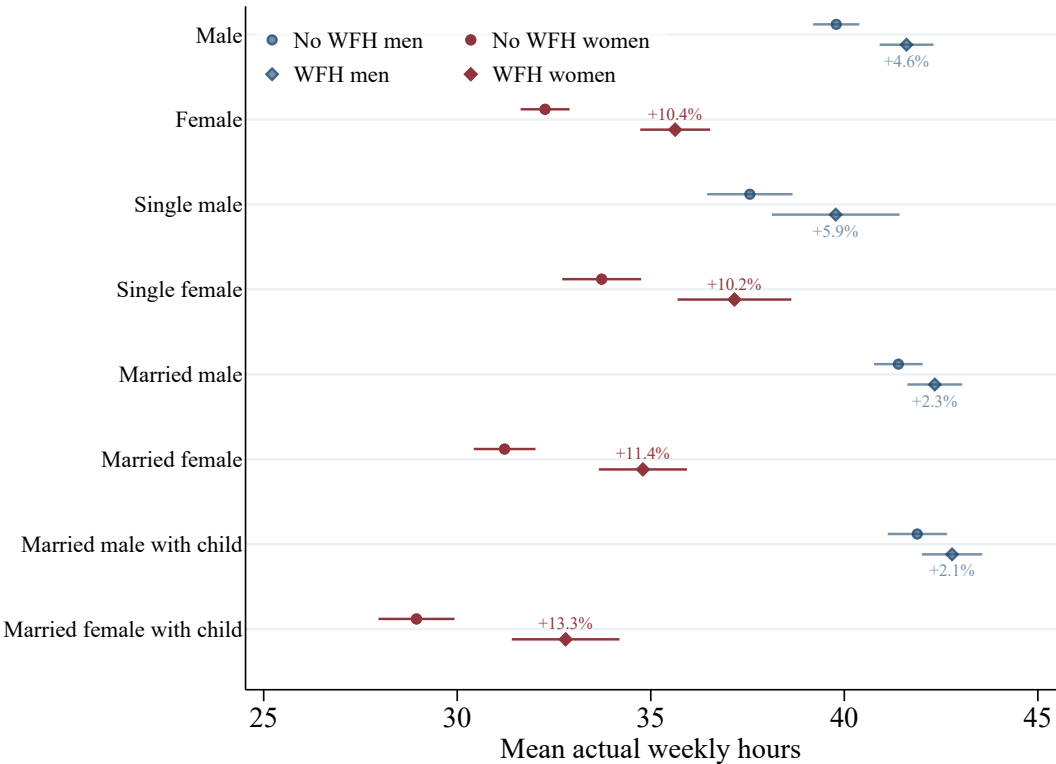
FIGURE 1. WFH intensity by group (2023)



Notes: Working-from-home intensity by demographic group in 2023. The figure reports mean working-from-home hours as a share of mean weekly working hours, separately for men and women across demographic groups. The sample is restricted as in the main analysis. We use survey weights to compute the statistics. Data source: SOEP 2023 (v40.1).

non-WFH individuals and diamonds denote WFH individuals, with the labels reporting the percentage difference in mean hours between the two. Across all groups, WFH is associated with higher usual hours, but the difference is consistently larger for women. In the full sample, women who work from home report 10.4 percent more hours than women who do not, compared with 4.6 percent for men. The same pattern appears among single individuals and is strongest among married individuals with children, where the gap is 13.3 percent for women and only 2.1 percent for men.⁷ Overall, the figure suggests that teleworking is associated with higher labour supply, especially for women in groups more likely to face stronger household and childcare burdens.

FIGURE 2. Group means of usual weekly hours (2023)



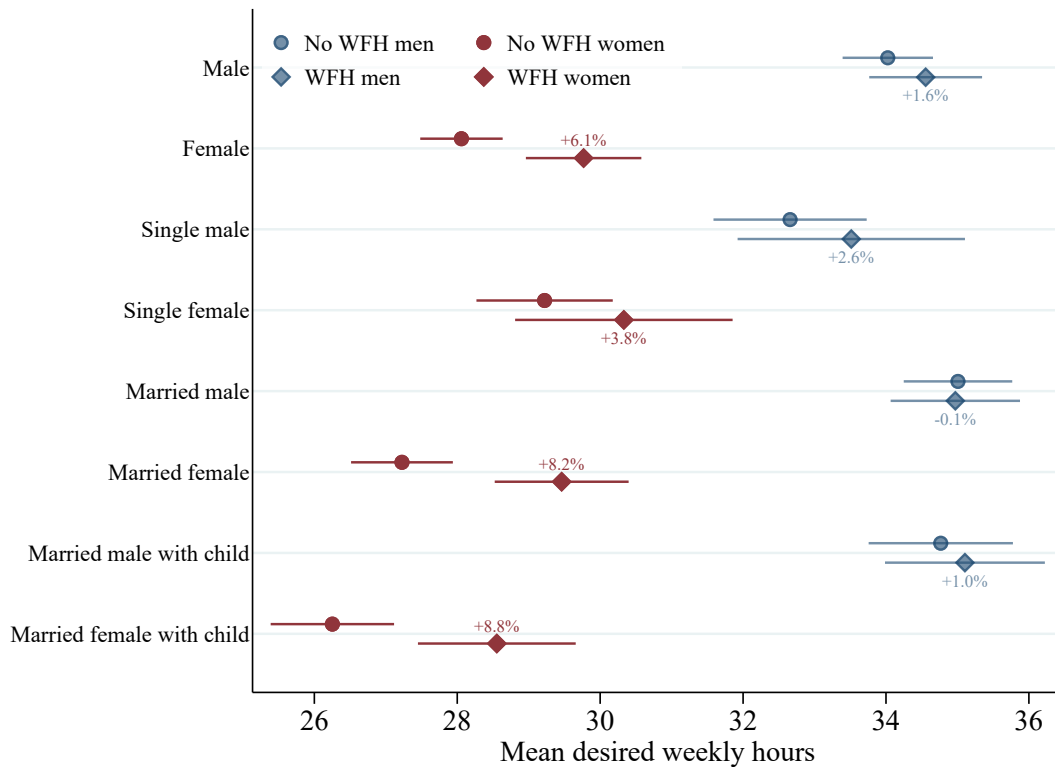
Notes: Usual weekly hours by work-from-home status and demographic group in 2023. The figure reports mean usual weekly hours and confidence intervals separately for men and women, distinguishing between workers who do not work from home and workers who do. Groups are defined by marital status and parental status. Labels next to the work-from-home observations report the percentage difference relative to the corresponding no-WFH group. The sample is restricted as in the main analysis. We use survey weights to compute the statistics. Data source: SOEP 2023 (v40.1).

We find a similar pattern when we focus on desired hours. Figure 3 shows that, for men, teleworking is associated with little or no difference in desired hours across demog-

⁷It is important to note that the group of single men in the sample is disproportionately smaller than the others, this increases the confidence interval of the estimates.

graphic groups, as the estimated gaps are generally small and not statistically significant. By contrast, for women, the association between WFH and desired hours is large and increasing in household responsibilities, going from 3.8 percent for single women, to 8.2 percent for married women and 8.8 percent for married women with children. Taken together, these results suggest that, for women, working from home is associated not only with a higher usual labor supply, but also with a higher desired labor supply. This pattern seems to support the interpretation that teleworking is valued as an amenity that relaxes time-coordination constraints and makes a larger labor supply more feasible, especially for women facing stronger home-production and childcare demands.

FIGURE 3. Group means of desired weekly hours (2023)



Notes: Desired weekly hours by work-from-home status and demographic group in 2023. The figure reports the weighted mean desired weekly hours and confidence intervals separately for men and women, distinguishing between workers who do not work from home and those who do. Groups are defined by marital status and parental status. Labels next to the work-from-home observations report the percentage difference relative to the corresponding no-WFH group. The sample is restricted as in the main analysis. We use survey weights to compute the statistics. Data source: SOEP 2023 (v40.1).

3. Empirical strategy and identification

As described earlier, we now explore the impact of WFH on usual and desired weekly hours of work using OLS and instrumental-variables (IV) estimators across multiple

specifications and sample restrictions.

We consider two complementary measures of remote work that capture the extensive and intensive margins of working from home. First, the extensive-margin treatment, $WFHdummy_i$, is a binary indicator equal to one if individual i worked from home at least once per week in 2023, and zero otherwise. This measure captures whether the worker participates in remote work. Second, the intensive-margin treatment, $WFHhours_i$, measures the intensity of remote work and is defined as the average number of hours per week that individual i works from home. This variable captures the degree to which remote work substitutes for workplace hours among individuals who engage in WFH.

3.1. Baseline OLS specification

Let $Y_i \in \{usual_hours, desired_hours\}$ denote the outcome of interest for individual i , and let D_i denote the individual's WFH arrangement, measured either at the extensive or intensive margin. Let $\mathbf{c}_i$ be a $k \times 1$ vector of observed individual characteristics and federal-state fixed effects.

We are interested in the causal effect of D_i on Y_i as summarized by the linear equation

$$(1) \quad Y_i = \beta_{ols} D_i + \mathbf{c}_i' \boldsymbol{\gamma}_{ols} + \varepsilon_i$$

where β_{ols} is the coefficient of interest, $\boldsymbol{\gamma}_{ols}$ is a $k \times 1$ vector of coefficients associated with the covariates, and ε_i is an individual-specific error term.

If D_i were exogenous conditional on $\mathbf{c}_i$, that is, if

$$(2) \quad \mathbb{E}(\varepsilon_i \mid D_i, \mathbf{c}_i) = 0,$$

then OLS would identify the average treatment effect (ATE), given by β_{ols} . In our setting, however, this condition is unlikely to hold. Even after controlling for a rich set of observed covariates and state fixed effects, working from home may remain correlated with unobserved determinants of labour-supply outcomes, such as job amenities, worker preferences, or household constraints. For this reason, we suspect that the estimator β_{ols} is biased. To overcome this problem, we rely on an IV strategy.

3.2. Instrumental-variables design

To address the endogeneity of D_i , we use the WFH feasibility index described in Section 2.2 as an instrument.

Let z_i denote the instrument, namely the pre-pandemic WFH-feasibility measure

assigned to individual i on the basis of their occupation. The 2SLS specification is given by

$$(3) \quad Y_i = \beta_{2sls} D_i + \mathbf{c}_i' \gamma_{2sls} + u_i,$$

$$(4) \quad D_i = \pi z_i + \mathbf{c}_i' \delta + v_i,$$

where Y_i is the outcome of interest, D_i denotes the WFH treatment variable, and $\mathbf{c}_i$ is a $k \times 1$ vector of controls. In the second-stage equation (Equation 3), β_{2sls} is the 2SLS coefficient of interest, measuring the effect of WFH on the outcome, γ_{2sls} is a $k \times 1$ vector of coefficients on the controls, and u_i is the second-stage error term. In the first-stage equation (Equation 4), π measures the partial effect of the instrument on WFH, δ is a $k \times 1$ vector of coefficients on the controls, and v_i is the first-stage error term.

The 2SLS estimator uses z_i as an instrument for D_i , controlling for $\mathbf{c}_i$ in both stages. We estimate this specification separately for each outcome and for both treatment definitions. Under the standard identifying assumptions, discussed below, the 2SLS estimator identifies an average of covariate-specific local average treatment effect (LATEs) in the sense of Angrist and Imbens (1995). In our setting, α is the causal effect of WFH for the subgroup of individuals whose working-from-home behavior is changed by exogenous variation in pre-pandemic WFH feasibility, namely the compliers.

3.3. Identifying assumptions

The 2SLS admits a causal interpretation under the standard assumptions of relevance, exogeneity, excludability, monotonicity and rich covariates (Angrist and Pischke 2009; Mogstad and Torgovitsky 2024). Next, we discuss these necessary conditions in our setting.

Relevance. In our setting, relevance means that pre-pandemic WFH feasibility must be correlated with post-pandemic WFH take-up, after partialling out the set observed covariates and federal-state fixed effects. The Cragg–Donald Wald statistics reported in Tables 2 and 3, as well as in tables C.1-C.16 in the Appendix, are uniformly very large and comfortably exceed conventional Stock–Yogo critical values. We therefore find no evidence of weak-instrument problems in any specification. Similar results are found in Alipour (2025) and Meister, Menkhoff, and Schröder (2025).

Exogeneity. Exogeneity requires that, conditional on the observed controls $\mathbf{c}_i$, the instrument z_i is unrelated to the unobserved determinants of the outcome. The exogeneity condition is

$$(5) \quad \mathbb{E}(u_i \mid z_i, \mathbf{c}_i) = \mathbb{E}(u_i \mid \mathbf{c}_i).$$

Equation (5) means that, once we condition on the included controls, variation in pre-pandemic WFH feasibility is not systematically related to unobserved factors affecting labour-supply outcomes.

This assumption is plausible in our setting because z_i is constructed from pre-pandemic occupation-based information and captures technological constraints on remote work rather than realized work arrangements or contemporaneous employer policies. A remaining concern is that workers may sort into jobs with different levels of WFH feasibility on the basis of unobserved preferences or labour-supply characteristics. To mitigate this concern, $\mathbf{c}_i$ includes a rich set of demographic and labour-market controls. Under this rich specification, the remaining variation in z_i is interpreted as reflecting occupation-based technological differences across jobs rather than unobserved determinants of Y_i .

Exclusion restriction. The exclusion restriction requires that the instrument affects the outcome only through its effect on the treatment D_i . Formally, for every treatment level and instrument value,

$$(6) \quad Y_i(D, z) = Y_i(D).$$

That is, conditional on treatment status, the instrument has no direct effect on the outcome. In our context, pre-pandemic WFH feasibility may affect usual or desired hours worked only by changing realized WFH arrangements. Intuitively, this assumption requires that the technological feasibility of WFH matters for hours outcomes only because it makes working from home more or less likely. Once realized WFH is held fixed, whether a job is more or less technically feasible to perform from home should not directly affect either the number of hours worked or the number of hours the individual would like to work.

This assumption is plausible in our setting for several reasons. First, the instrument is constructed from a pre-pandemic assessment of WFH feasibility. It therefore captures technological constraints on remote work rather than realized work arrangements or contemporaneous employer policies in 2023. Second, the measure is defined at the level of occupation content, so its natural interpretation is precisely as a shifter of the opportunity to work from home, not as a direct determinant of hours preferences or hours supplied. Third, our specifications condition on a rich set of demographic and labour-market controls, which helps absorb systematic differences in worker composition across jobs with different levels of WFH feasibility.

A natural concern is that even in jobs with low WFH feasibility, the feasibility measure may proxy for other job characteristics that directly affect labour-supply outcomes, such as job flexibility, autonomy, workload, or career concerns. To address it, in Section 5.1 we implement a robustness check designed to probe whether the instrument appears to operate through channels other than realized WFH.

Monotonicity. For the LATE interpretation we assume monotonicity, meaning that higher WFH feasibility increases the probability of working from home in the extensive case or increases the number of hours worked from home, in the intensive case. In other words, we assume the absence of “defiers” in the population. This assumption is natural in the present context because greater occupation-based feasibility expands the set of jobs that can technically be performed remotely.

Rich covariates. Following [Blandhol et al. \(2022\)](#), our identification strategy relies on conditioning on a sufficiently rich set of observed covariates. Let $L(z_i | \mathbf{c}_i)$ denote the linear projection of the instrument z_i onto the linear span of $\mathbf{c}_i$. The rich-covariates condition requires

$$(7) \quad \mathbb{E}(z_i | \mathbf{c}_i) = L(z_i | \mathbf{c}_i).$$

Equation (7) requires that the dependence of the instrument on the observed covariates be fully captured by its linear projection on the included controls. Intuitively, once we condition on a sufficiently rich set of demographic, household, and labour-market characteristics, the remaining variation in pre-pandemic WFH feasibility should reflect technological occupation-based differences across jobs rather than worker sorting or other omitted determinants of labour-supply outcomes.

3.4. Heterogeneous effects assessment

We also estimate all specifications separately for women and men in order to examine potential heterogeneity in the effects of working from home on hours of work. As discussed above, labour-supply preferences, realized working hours, and the allocation of household responsibilities differ systematically by gender. In particular, women may be more likely to adjust their labour supply in response to childcare needs and other family constraints, which may in turn interact with the flexibility provided by remote work.

An alternative approach would be to estimate a pooled specification including an interaction between WFH and a gender indicator. In the IV setting, however, this would require treating both D_i and $D_i \times \text{Women}_i$ as endogenous, where Women_i is an indicator for being a woman. This, in turn, would require at least one additional valid excluded instrument for the interaction term. Since we only have one excluded instrument, such a specification would be underidentified. Estimating the model separately by gender therefore provides a simple and transparent way to investigate heterogeneity in the causal effect of WFH.

4. Main results

We begin with the full-sample estimates for usual and desired weekly hours, reported in Table 2. The detailed regression outputs are reported in Appendix Tables C.1 to C.4. Throughout, we consider the two treatment definitions. All specifications are estimated under three progressively richer control sets. Specification (1) includes age, age squared, a female indicator, and federal state fixed effects. Specification (2) adds education and labor-market experience controls. Specification (3) further adds marital status, the presence of a child below the age of 12, and interactions between these household variables and the female indicator. Since the credibility of the instrument depends on conditioning on a sufficiently rich set of household and labor-supply characteristics, we treat specification (S.3) as our preferred specification.

The full-sample estimates indicate a positive effect of working from home on both usual and desired hours. In the extensive-margin specification, the OLS coefficient in (3) is 1.866 for usual hours and 0.819 for desired hours, while the corresponding 2SLS estimates are 7.282 and 3.552. In the intensive-margin specification, the OLS coefficients in (3) are 0.128 for usual hours and 0.075 for desired hours, compared with 2SLS estimates of 0.315 and 0.154. Across both treatment definitions working from home increases realized labor supply and also raises the number of hours workers report wishing to supply.

A further feature of the results is that the effect on usual hours is consistently larger than the effect on desired hours. This suggests that, although WFH expands desired labor supply, the increase in realized hours is even more pronounced. One possible interpretation is that remote work reduces the frictions that ordinarily reduce realized hours. In particular, the absence of commuting may free time that is partly reallocated to work, thereby extending usual working time beyond what would be observed under on-site arrangements. In addition, remote work may weaken the informal coordination of the end of the workday that is more common in on-site settings, where workers observe colleagues leaving and face clearer social pressure about when work has ended. Under WFH, these cues are less salient, which may make workers more likely to continue working beyond their usual stopping point. Taken together, these mechanisms are consistent with a setting in which WFH increases both desired and usual hours, but has a larger effect on the latter.

A noticeable feature of the results is that the 2SLS estimates are consistently larger than the corresponding OLS coefficients. This pattern is consistent with OLS understating the causal effect of WFH, under the maintained validity of the instrument. One plausible interpretation is negative selection into remote work. Workers who sort into, or are sorted into, WFH arrangements may also face unobserved constraints that depress labor supply, such as family responsibilities or other competing demands on time. In that case, the OLS coefficient reflects both the positive effect of WFH and negative selection on unobservables,

biasing the estimated relationship downward. Once endogeneity is addressed using pre-pandemic WFH feasibility as an instrument, the estimated effect becomes materially larger.

TABLE 2. Main results: full sample

	Weekly actual hours			Weekly desired hours		
	(1)	(2)	(3)	(1)	(2)	(3)
<i>Extensive margin: WFH dummy</i>						
OLS	2.954*** (0.384)	1.840*** (0.414)	1.866*** (0.406)	1.651*** (0.414)	0.864* (0.449)	0.819* (0.447)
2SLS	8.379*** (0.884)	6.964*** (1.343)	7.282*** (1.335)	4.830*** (0.906)	3.442** (1.417)	3.552** (1.417)
Cragg–Donald Wald <i>F</i>	767.75	334.53	331.14	767.75	334.53	331.14
<i>Intensive margin: WFH hours</i>						
OLS	0.164*** (0.013)	0.130*** (0.014)	0.128*** (0.014)	0.101*** (0.015)	0.078*** (0.016)	0.075*** (0.016)
2SLS	0.369*** (0.039)	0.302*** (0.058)	0.315*** (0.057)	0.212*** (0.040)	0.149** (0.061)	0.154** (0.061)
Cragg–Donald Wald <i>F</i>	574.32	255.50	253.81	574.32	255.50	253.81
Observations	2,627	2,627	2,627	2,627	2,627	2,627
Age, gender, state FE	yes	yes	yes	yes	yes	yes
Education & experience	no	yes	yes	no	yes	yes
Family controls & interactions	no	no	yes	no	no	yes

Notes: The table reports the estimated effect of working from home on usual and desired weekly hours worked. In the extensive-margin specifications, the treatment is the WFH dummy. In the intensive-margin specifications, the treatment is weekly hours worked from home. OLS rows report ordinary least squares estimates. 2SLS rows report instrumental-variables estimates using WFH feasibility as an instrument. Robust standard errors are reported in parentheses. Specification (1) includes age, age squared, a gender indicator, and federal-state fixed effects. Specification (2) adds education and labour-market experience controls. Specification (3) further adds family controls and interactions. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.1. Women versus men

The gender split, reported in Table 3, reveals marked heterogeneity in the effects of working from home. The detailed regression outputs are reported in Appendix Tables C.13–C.16. For women, WFH increases both usual and desired hours, and it does so at both the extensive and intensive margins. In specification (3), the extensive-margin 2SLS coefficient is 10.923 for usual hours and 7.248 for desired hours. At the intensive margin, the corresponding coefficients are 0.530 for usual hours and 0.352 for desired hours. These are economically large effects, and in each case they lie well above the corresponding OLS estimates. For women, remote work therefore does not merely increase realized hours, it also raises the number of hours they would like to work. This pattern is consistent with the view that WFH relaxes constraints that would otherwise limit female labor supply.

The pattern for men is quite different. WFH continues to have a positive effect on usual hours. In specification (3), the extensive-margin 2SLS coefficient for usual hours is 3.597, while the intensive-margin coefficient is 0.141. By contrast, the corresponding

coefficients for desired hours are statistically insignificant. Thus, for men, WFH appears to increase realized labor supply without materially changing the number of hours they would prefer to work.

This contrast is informative. For men, remote work seems to affect the organization and timing of work rather than underlying labor-supply preferences. One possible interpretation is that WFH makes it easier for work to spill-over beyond standard working hours, thereby increasing usual hours without inducing a parallel rise in desired hours. In other words, for men, remote work may change how work is performed and distributed over the day, but not the amount of work they would ideally choose to supply.

For women, by contrast, the fact that both usual and desired hours rise suggests a different mechanism. Here, WFH appears not only to increase realized labor supply, but also to expand the set of work arrangements that women regard as feasible or attractive. This is consistent with the idea that remote work relaxes constraints linked to commuting, schedule rigidity, and the allocation of time across market and non-market activities. More broadly, it suggests that WFH may strengthen women's attachment to the labor market by making more intensive work arrangements easier to sustain.

Taken together, the results in Table 3 suggest that the labor-supply effects of remote work are concentrated among women. For women, WFH appears to relax constraints that affect both the ability to work and the desired amount of work. For men, by contrast, the evidence points mainly to an effect on realized hours, with little indication that WFH changes preferred labor supply. This asymmetry is consistent with the broader view that remote work is especially valuable when labor supply is limited by time-allocation frictions, household constraints, or barriers to maintaining a stronger attachment to market work.

4.2. Full-time versus part-time workers

One concern with the gender results presented above is that they may partly reflect differences in the scope for hours adjustment rather than gender per se. Men in our sample are predominantly full-time workers (91%), leaving little room to increase hours further, while women are more likely to work part-time (53%). To assess this, we re-estimate the main specifications separately for full-time and part-time workers.

The results are reported in Appendix Tables C.5–C.12. For full-time workers (Tables C.5–C.8), none of the 2SLS estimates in specification (3) is statistically significant, either at the extensive or intensive margin, for usual or desired hours. For part-time workers (Tables C.9–C.12), by contrast, the effects are large and precisely estimated. The extensive-margin 2SLS coefficient is 10.282 for usual hours and 4.508 for desired hours. At the intensive margin, the corresponding coefficients are 0.628 for usual hours and 0.275 for desired hours, all well above the corresponding OLS estimates.

TABLE 3. Gender heterogeneity: usual and desired weekly hours

	Weekly actual hours		Weekly desired hours	
	Men	Women	Men	Women
<i>Extensive margin: WFH dummy</i>				
OLS	0.919* (0.515)	2.825*** (0.609)	0.132 (0.668)	1.416** (0.607)
2SLS	3.597** (1.817)	10.923*** (1.945)	−0.478 (2.322)	7.248*** (1.829)
Relative value	8.7%	32.3%	−1.4%	25.5%
Cragg–Donald Wald <i>F</i>	131.19	184.09	131.19	184.09
<i>Intensive margin: WFH hours</i>				
OLS	0.063*** (0.016)	0.204*** (0.022)	0.041* (0.021)	0.112*** (0.024)
2SLS	0.141** (0.070)	0.530*** (0.095)	−0.019 (0.091)	0.352*** (0.089)
Cragg–Donald Wald <i>F</i>	108.25	127.62	108.25	127.62
Observations	1,254	1,373	1,254	1,373

Notes: The table reports gender-specific estimates of the effect of working from home on usual and desired weekly hours worked. In the extensive-margin specifications, the treatment is the WFH dummy. In the intensive-margin specifications, the treatment is weekly hours worked from home. OLS rows report ordinary least squares estimates. 2SLS rows report instrumental-variables estimates using WFH feasibility as an instrument. Robust standard errors are reported in parentheses. Relative values report the 2SLS coefficient as a percentage of the corresponding gender-specific mean of the dependent variable. All specifications use the full control set: age, age squared, federal-state fixed effects, education and labour-market experience controls, family controls, and interactions. Cragg–Donald Wald *F* statistics are reported for the 2SLS specifications. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

These findings are consistent with a ceiling-effect interpretation for full-time workers, but they do not undermine the gender result. Rather, the two sets of results reflect the same underlying mechanism: WFH expands labour supply where there is scope for adjustment, and women’s higher incidence of part-time employment is itself a symptom of the time-coordination constraints that WFH relaxes. Moreover, a pure ceiling-effect story cannot fully account for the pattern observed among men: if men were simply constrained by a contractual or practical upper bound on hours, WFH should leave usual hours unchanged as well. Instead, we find that WFH significantly increases usual hours for men, while leaving desired hours unaffected. This pattern is more consistent with a work-intensification mechanism than with a ceiling effect, and supports the interpretation that the gender asymmetry in desired hours reflects a difference in the constraints that remote work relaxes. The part-time and gender results are therefore complementary, meaning that part-time status is a mediator through which the flexibility channel operates, not an alternative explanation for it.

5. Robustness exercises

This section reports additional checks designed to assess the credibility of the identification strategy.

5.1. Placebo robustness for excludability

Tables C.17 and C.18 report placebo regressions for workers in the bottom 10% and bottom 25% of the 2019 WFH-feasibility distribution. For these workers, the technical scope for working from home is very limited by construction. Hence, if the instrument affects labour-supply outcomes only through the opportunity to work from home, its relationship with usual and desired hours should become negligible in these low-feasibility groups.

The results are consistent with this prediction. In both Table C.17 and Table C.18, the IV estimates are extremely imprecise and statistically uninformative in the bottom 10% and bottom 25% placebo samples. For example, in the bottom 10% sample, the extensive-margin IV coefficients are 121.852 for usual hours, 586.287 for desired hours, all with very large standard errors. The corresponding intensive-margin estimates are 1.561, 7.511, and 4.460, again with very large standard errors. The same pattern holds in the bottom 25% sample.

Importantly, these placebo regressions are accompanied by an extremely weak first stage. The Cragg–Donald Wald statistics are close to zero throughout, ranging from 0.04 to 0.52 across specifications, and the Anderson–Rubin p-values are uniformly far from conventional significance levels. This is what one would expect if, among workers whose tasks are barely compatible with remote work, WFH feasibility has little power to shift

actual WFH behaviour.

Taken together, Tables C.17 and C.18 provide supportive placebo evidence for the exclusion restriction. Once the sample is restricted to workers at the bottom of the WFH-feasibility distribution, there is no evidence of a systematic effect on usual hours or desired hours. Since the first stage is extremely weak in these placebo samples, however, the exercise should be interpreted as supportive rather than definitive evidence.

5.2. Rich-covariates robustness

Tables C.19 and C.20 compare the baseline IV estimates with the corresponding instrumental variable residual (IVR) estimates for usual and desired hours worked. The purpose of this exercise is to assess whether the main findings are sensitive to the rich-covariates condition. As discussed above, when covariates are included, the standard 2SLS estimand recovers the causal parameter of interest only if this condition holds. The IVR estimator proposed by [Glorias, Martellosio, and Santos Silva \(2025\)](#) addresses this concern by residualizing the instrument with respect to the covariates. Since the residualized instrument has conditional mean zero given the covariates, the rich-covariates condition is satisfied by construction.⁸

The results show that using IVR makes essentially no difference to the estimated treatment effects. For usual hours worked, Table C.19 shows that, in the full sample, the extensive-margin estimate changes only from 7.282 under baseline IV to 7.309 under IVR, while the intensive-margin estimate changes from 0.315 to 0.318. Among women, the corresponding estimates move from 10.923 to 11.009 and from 0.530 to 0.536, respectively. For desired hours worked, Table C.20 shows the same pattern: in the full sample, the extensive-margin estimate changes from 3.552 to 3.612, while the intensive-margin estimate moves from 0.154 to 0.157. Among women, the corresponding estimates change from 7.248 to 7.330 and from 0.352 to 0.357.

Overall, Tables C.19 and C.20 indicate that the main findings are not driven by a failure of the rich-covariates condition. The baseline results are therefore not sensitive to this particular concern.

5.3. Unrestricted-sample robustness

Tables C.21 and C.22 report a robustness exercise in which we re-estimate the main specifications using an unrestricted sample that allows individuals to have changed jobs between 2019 and 2023. The baseline analysis restricts the sample to job stayers in order

⁸We follow the residualization procedure proposed by [Glorias, Martellosio, and Santos Silva \(2025\)](#). Specifically, we construct

$$z_i^* = WFHFeas_i - \mathbb{E}[WFHFeas_i \mid \mathbf{c}_i],$$

where $\mathbb{E}[WFHFeas_i \mid \mathbf{c}_i]$ is estimated using a shallow neural network.

to limit concerns that workers may have sorted into more remote-workable jobs precisely because they valued the WFH amenity. This restriction is therefore conservative, but it may also exclude workers whose job mobility is itself part of the broader adjustment to remote-work opportunities.

The results are highly stable once this restriction is removed. For usual hours worked, Table C.21 shows that the full-sample estimates remain positive and statistically significant at both the extensive and intensive margins. The same pattern holds for part-time workers and women, for whom the estimated effects remain particularly large. For men, the extensive-margin effect on usual hours remains positive, while the intensive-margin effect is weaker and not statistically significant, matching the qualitative pattern found in the baseline sample. Note also that the changes in the magnitudes of the estimates are also residual.

Table C.22 shows a similarly stable pattern for desired hours worked. The estimated effects remain positive in the full sample and are strongest among part-time workers and women. By contrast, the estimates for men remain statistically insignificant at both margins. Once again, the changes in magnitude of the estimated values are small. Overall, the unrestricted-sample results therefore confirm that the main findings do not depend on the job-stayer restriction. Although that restriction is useful as a precaution against endogenous job sorting, relaxing it does not materially alter the empirical conclusions.

6. Conclusion

This paper provides a first attempt to analyze the causal effects of remote work on household labor supply in Germany. To investigate this question, we use German SOEP data and focus on a sample of prime-age individuals who did not change jobs between 2019 and 2023. After showing that the adoption of working from home is associated with higher usual and desired labor supply, especially among women, we move to a more in-depth analysis.

To identify more precisely the effects of WFH on labor supply, we adopt an IV identification strategy. Specifically, we study the effect of an additional hour of WFH on both usual working hours and desired working hours. To address endogeneity concerns related to workers' adoption of WFH, we instrument weekly WFH hours with an occupation-level index of WFH feasibility computed prior to the pandemic. This index allows us to purge the weekly WFH hours variable of its endogenous component.

Regarding usual hours, we find that WFH increases working hours for both men and women, whereas the effects are gender asymmetric when it comes to desired hours. In the full sample, working from home at least once per week increases usual weekly hours by around 7.3 hours and desired weekly hours by around 3.6 hours. However, this average effect masks substantial heterogeneity across gender. For women, the estimated effects

are large both on usual and desired labor supply: WFH increases usual hours by around 11 hours per week and desired hours by more than 7 hours per week. By contrast, for men, WFH increases usual working hours but has no statistically significant effect on desired hours.

Taken together, these findings provide direct evidence supporting the *flexibility channel* discussed in the introduction. The fact that WFH raises not only realized but also the "desired" labor supply for women suggests that remote work relaxes time-coordination frictions between market work and home production, expanding the feasible set of labor supply choices for constrained workers. For men, by contrast, the increase in usual hours, combined with unchanged desired hours, appears more consistent with a work reorganization or work-intensification mechanism.

Overall, our results suggest that remote work may represent an important margin through which labor supply constraints, especially for women, can be attenuated. This is particularly relevant in aging economies characterized by labor shortages and persistently low female labor force participation. At the same time, our analysis remains only a first step. Future work should further investigate the mechanisms behind these gender asymmetries, explore heterogeneity across family structures, income groups, and contractual arrangements, and study the long-run implications of remote work for gender inequality and household labor allocation.

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Online Appendix

Appendix A. Data

A.1. Variables of interest

Employment status: Respondents with positive net monthly earnings are classified as employed.

Usual weekly hours of work: Employed respondents report both contractual and usual (“actual” is SOEP terminology) weekly working hours. Usual hours are contractual hours plus any paid and unpaid overtime. Hours of work are winsorised at the 1st and 99th percentiles.

Desired hours of work: Employed respondents are asked: “If you could choose your own working hours, taking into account that your income would change according to the number of hours: How many hours would you want to work?” Desired hours are winsorised at the 1st and 99th percentiles.

Work experience: Work experience reflects the total length of full-time and part-time employment in the respondent’s career up to the point of the interview.

Job changes: This variable captures a change of job since the previous interview, including transitions between and within a firms.

WFH: We construct two measures of working from home. The *WFH dummy* equals one if the respondent reports having worked from home at least once per week on average over the past six months, and zero otherwise, including respondents who report never having worked from home. *WFH hours* is the average number of hours per week the respondent typically works from home. In the small number of cases where reported WFH hours exceed usual working hours, we cap WFH hours at usual hours worked.

Socio-demographics: Socio-demographic characteristics include migration background, years of schooling, university degree attainment, marital and partnership status, and the birth year of each child and the number of children under 18 living in the household.

A.2. Sample construction

Table A.1 reports the number of individuals remaining at each stage of the sample selection. We begin with all SOEP respondents observed at least once between 2019 and 2023. We then retain only individuals interviewed in every wave over this period, so that job-change status can be tracked consistently from the pre-pandemic baseline. Among these, we drop individuals who report a job change (either within the same firm as well as a change to another firm) in any wave after 2019; this ensures that the pre-pandemic WFH feasibility index, which is assigned at the occupational level, reflects the occupation held throughout the entire period rather than a post-pandemic reallocation. Turning to the cross-section,

we restrict attention to the 2023 wave and retain individuals aged 18-65 who are employed, excluding the self-employed. We then drop observations with missing values for our key outcome variables (usual and desired weekly hours of work), the occupational category, the WFH variables (the WFH dummy, weekly WFH hours, and the feasibility index), and the set of control variables (age, federal state, years of education, university degree, work experience, marital status). Finally, we exclude a small number of implausible observations: full-time workers reporting fewer than 30 weekly usual hours.

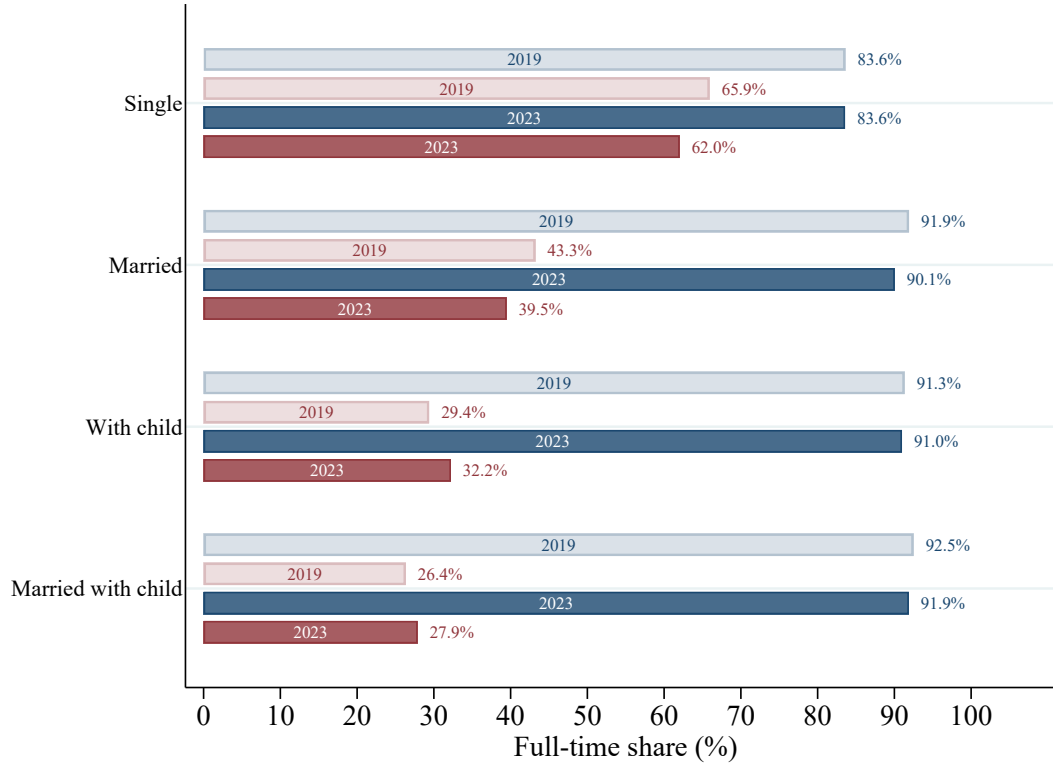
TABLE A.1. Sample construction

Sample restriction	Individuals
<i>Panel restrictions (2019–2023)</i>	
SOEP respondents observed in 2019–2023	111,514
Observed in all five waves	39,579
No job change since 2019	34,337
<i>Cross-section restrictions (2023)</i>	
Aged 18–65	20,375
Employed	5,286
Excluding self-employed	4,633
Non-missing actual hours	4,351
Non-missing desired hours	4,242
Non-missing occupational category	2,861
Non-missing WFH variables	2,750
Non-missing control variables	2,689
Hours plausibility restrictions	2,627
Final sample	2,627
of which women	1,373
of which men	1,254

Notes: Each row reports the number of unique individuals remaining after the listed restriction is applied cumulatively. Panel restrictions are applied over the 2019–2023 period; cross-section restrictions are applied to the 2023 wave. WFH variables comprise the WFH dummy, weekly WFH hours, and the WFH feasibility index. Control variables comprise age, federal state, years of schooling, university degree, labour market experience, and marital status. Hours plausibility restrictions exclude full-time workers reporting fewer than 30 weekly working hours and individuals reporting zero usual or desired hours. Data source: SOEP 2023 (v40.1).

Appendix B. Additional Graphs

FIGURE B.1. Full-time share by category, sex, and year



Notes: Full-time share by category, sex, and year. The figure reports the weighted share of individuals working full-time in 2019 and 2023 for four demographic groups: single, married, with child, and married with child. Bars are shown separately for men and women. Lighter bars correspond to 2019 and solid bars to 2023. The sample includes individuals aged 25-55, excludes the self-employed, and applies the same sample restrictions used in the main analysis.

These demographic groups can be interpreted as capturing different degrees of household time constraints. Single individuals are, on average, less likely to face substantial childcare and household responsibilities. By contrast, cohabiting individuals are more likely to share a household structure that entails greater coordination needs, while the presence of children plausibly increases these constraints further. Such demands are likely to be strongest, on average, among cohabiting couples with at least one child under the age of 18.

A first pattern, consistent with the existing literature, is that full-time employment is substantially more common among men than among women. Moreover, this gender gap widens within cohabiting households, a pattern suggestive of intra-household specialization in labor supply. Turning to the evolution over time, the group of single individuals displays relative stability for men: the share working full-time is identical in 2019 and 2023,

at 83.6 percent. Among single women, instead, the full-time share declines modestly, from 65.9 to 62.0 percent. A similar pattern emerges among cohabiting individuals without children: men remain broadly stable, with the full-time share moving only from 91.9 to 90.1 percent, whereas for women it declines from 43.3 to 39.5 percent.

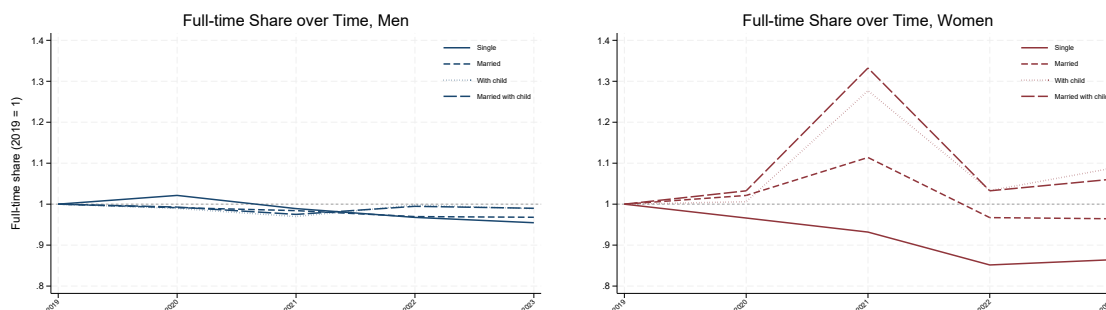
The picture changes when we consider only cases where children are present in the household. Among individuals living with at least one child under 18, the male full-time share remains essentially unchanged, moving from 91.3 to 91.0 percent, while the female full-time share rises from 29.4 to 32.2 percent, an increase of about 10 percent relative to its 2019 level. A similar pattern is observed among cohabiting individuals with children: the male full-time share is again stable, declining only slightly from 92.5 to 91.9 percent, whereas the female share increases from 26.4 to 27.9 percent, corresponding to an increase of about 6 percent relative to 2019. Overall, the groups in which women are more likely to face stronger childcare and household responsibilities are precisely those in which female full-time employment increases in the post-pandemic period, consistent with the hypothesis that the spread of teleworking relaxed time-coordination constraints and facilitated labor supply at the intensive margin.

However, the comparison in Figure 1 is based on only two points in time and therefore cannot fully capture the dynamics of adjustment over the post-pandemic period. Figure 2 provides a broader perspective by plotting, from 2019 to 2023, the share of full-time workers within each demographic group normalized to 1 in 2019. The left panel reports the evolution for men, while the right panel reports the corresponding patterns for women. This normalization facilitates the comparison between the pre-pandemic period and the years following the diffusion of teleworking.

A first striking pattern is a clear gender asymmetry. For men, the share of full-time employment remains remarkably stable throughout the entire period and across all demographic groups. By contrast, women display substantial variation, with the strongest increases concentrated among the groups that are more likely, on average, to bear greater childcare and household responsibilities. In particular, women in cohabiting couples with at least one child - represented by the long-dashed line - in the household experience the largest increase in full-time employment relative to 2019. For this group, the full-time share rises by more than 30 percent at its peak in 2021, before stabilizing in 2023 at a level still about 6 percent above its pre-pandemic value. A similar pattern emerges for women with children more generally (the dotted line), whose full-time employment also increases sharply during the pandemic years and remains above its 2019 level in 2023. Married women (short-dashed line) without children display a more moderate but still visible increase during the same period, followed by stabilization at a level above the pre-pandemic baseline. By contrast, single women (solid line) show no comparable expansion and instead exhibit a persistent decline relative to 2019.

Taken together, these patterns suggest that the spread of remote work may have relaxed time-coordination constraints in ways that disproportionately facilitated an expansion of labor supply at the intensive margin for women, particularly for those facing greater home-production responsibilities.

FIGURE B.2. Evolution of the full-time share by demographic group and sex



Notes: Evolution of the full-time share by demographic group and sex, normalized to 1 in 2019. The left panel reports men and the right panel women. Each line plots the weighted share of individuals working full-time for one demographic group—single, married, with child, and married with child—normalized by its 2019 value. Values above one indicate an increase in the full-time share relative to 2019. The sample is restricted as in the main analysis. Data source: SOEP 2023 (v40.1).

At this point, it becomes natural to examine the diffusion of working from home across the demographic groups considered above. If teleworking alleviates the burden of home production and childcare by improving time coordination, then the groups that appear to benefit the most in terms of labor supply should also be those making more intensive use of this work arrangement. Figure 3 addresses this point by reporting, for each demographic group and separately by gender, the intensity of WFH in 2023. We measure WFH intensity within group g as the ratio between average weekly hours worked from home and average total weekly working hours:

$$\text{WFH Intensity}_g = \frac{\overline{\text{WFH hours}_g}}{\overline{\text{weekly hours}_g}} \times 100.$$

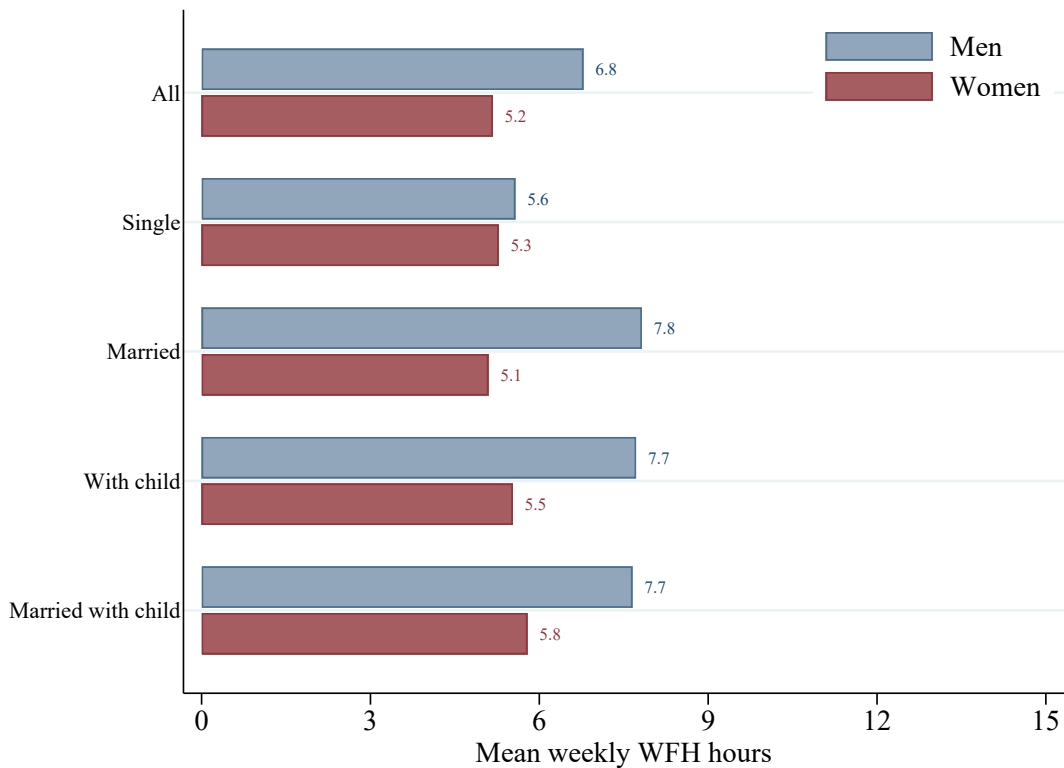
This measure captures the share of weekly work time that is performed from home within each demographic group.

Looking first at all employed prime-age individuals, WFH intensity is somewhat higher for men than for women: men work on average 16.9 percent of their weekly hours from home, compared with 15.1 percent for women. This aggregate gender difference likely reflects composition effects, as women are disproportionately employed in occupations with fewer teleworking opportunities and a greater incidence of contact-intensive tasks. Across demographic groups, however, the picture is more nuanced. Among single individuals, women display slightly higher WFH intensity than men (14.8 versus 14.3 percent).

Among married individuals without children, the pattern reverses sharply: men work 18.9 percent of their weekly hours from home, compared with 15.2 percent for women, a gap consistent with more traditional labor division and specialization within couples.

The most remarkable pattern emerges once children are present in the household. Among women with children, WFH intensity rises to 17.2 percent, and among married women with children it reaches 18.7 percent, the highest value observed for women across all groups. In other words, the women who are most likely to face stronger household and childcare responsibilities are also those who make the most intensive use of teleworking. This is consistent with the idea that WFH operates as a flexibility margin that is particularly valuable when the need to coordinate market work with home production is stronger.

FIGURE B.3. Average weekly WFH hours by demographic group (2023)



Notes: Average weekly hours worked from home by demographic group in 2023. The figure shows the average work-from-home hours for men and women across the following groups: all, single, married, with child, and married with child. The sample is restricted as in the main analysis. Data source: SOEP 2023 (v40.1).

Appendix C. Results Tables

TABLE C.1. Extensive Effect of WFH on Usual Hours Worked

	(1)	(2)	(3)
<i>Panel A. OLS</i>			
WFH dummy	2.954*** (0.384)	1.840*** (0.414)	1.866*** (0.406)
Women	-7.618*** (0.359)	-6.623*** (0.358)	-4.053*** (0.544)
College		2.312*** (0.584)	2.481*** (0.574)
Women × Kid			-2.202** (0.881)
Women × Married			-4.390*** (0.726)
Education and experience controls	no	yes	yes
Family controls and interactions	no	no	yes
Observations	2627	2627	2627
R ²	0.173	0.218	0.250
<i>Panel B. IV (2SLS)</i>			
WFH dummy	8.379*** (0.884)	6.964*** (1.343)	7.282*** (1.335)
Women	-7.380*** (0.372)	-6.513*** (0.370)	-4.170*** (0.557)
College		1.594** (0.628)	1.751*** (0.621)
Women × Kid			-2.537*** (0.928)
Women × Married			-3.911*** (0.750)
Education and experience controls	no	yes	yes
Family controls and interactions	no	no	yes
Observations	2627	2627	2627
Cragg-Donald Wald F	767.75	334.53	331.14

Notes: The dependent variable is usual weekly hours worked. Robust standard errors are reported in parentheses. * p<0.10, ** p<0.05, *** p<0.01. In Panel B, *WFH dummy* is instrumented with *WFH feasibility*. All specifications include age, age squared, a gender indicator, and federal-state fixed effects.

TABLE C.2. Extensive Effect of WFH on Desired Hours Worked

	(1)	(2)	(3)
<i>Panel A. OLS</i>			
WFH dummy	1.651*** (0.414)	0.864* (0.449)	0.819* (0.447)
Women	-5.915*** (0.389)	-5.449*** (0.399)	-3.474*** (0.645)
College		1.314** (0.637)	1.388** (0.634)
Women × Kid			0.550 (0.958)
Women × Married			-3.989*** (0.805)
Education and experience controls	no	yes	yes
Family controls and interactions	no	no	yes
Observations	2627	2627	2627
R ²	0.108	0.120	0.133
<i>Panel B. IV (2SLS)</i>			
WFH dummy	4.830*** (0.906)	3.442** (1.417)	3.552** (1.417)
Women	-5.775*** (0.394)	-5.393*** (0.402)	-3.533*** (0.646)
College		0.952 (0.670)	1.019 (0.667)
Women × Kid			0.380 (0.967)
Women × Married			-3.747*** (0.814)
Education and experience controls	no	yes	yes
Family controls and interactions	no	no	yes
Observations	2627	2627	2627
Cragg-Donald Wald F	767.75	334.53	331.14

Notes: The dependent variable is desired weekly hours worked. Robust standard errors are reported in parentheses. * p<0.10, ** p<0.05, *** p<0.01. In Panel B, *WFH dummy* is instrumented with *WFH feasibility*. All specifications include age, age squared, a gender indicator, and federal-state fixed effects.

TABLE C.3. Intensive Effect of WFH on Usual Hours Worked

	(1)	(2)	(3)
<i>Panel A. OLS</i>			
WFH hours	0.164*** (0.013)	0.130*** (0.014)	0.128*** (0.014)
Women	-7.400*** (0.356)	-6.456*** (0.355)	-3.992*** (0.543)
College		2.129*** (0.577)	2.314*** (0.566)
Women $\times$ Kid			-2.222** (0.871)
Women $\times$ Married			-4.204*** (0.720)
Education and experience controls	no	yes	yes
Family controls and interactions	no	no	yes
Observations	2627	2627	2627
R^2	0.191	0.232	0.263
<i>Panel B. IV (2SLS)</i>			
WFH hours	0.369*** (0.039)	0.302*** (0.058)	0.315*** (0.057)
Women	-6.965*** (0.371)	-6.185*** (0.375)	-3.962*** (0.556)
College		1.549** (0.622)	1.700*** (0.615)
Women $\times$ Kid			-2.420*** (0.904)
Women $\times$ Married			-3.689*** (0.745)
Education and experience controls	no	yes	yes
Family controls and interactions	no	no	yes
Observations	2627	2627	2627
Cragg-Donald Wald F	574.32	255.50	253.81

Notes: The dependent variable is usual weekly hours worked. Robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. In Panel B, *WFH hours* is instrumented with *WFH feasibility*. All specifications include age, age squared, a gender indicator, and federal-state fixed effects.

TABLE C.4. Intensive Effect of WFH on Desired Hours Worked

	(1)	(2)	(3)
<i>Panel A. OLS</i>			
WFH hours	0.101*** (0.015)	0.078*** (0.016)	0.075*** (0.016)
Women	-5.773*** (0.390)	-5.343*** (0.400)	-3.444*** (0.644)
College		1.170* (0.634)	1.253** (0.632)
Women × Kid			0.521 (0.955)
Women × Married			-3.856*** (0.803)
Education and experience controls	no	yes	yes
Family controls and interactions	no	no	yes
Observations	2627	2627	2627
R ²	0.115	0.126	0.138
<i>Panel B. IV (2SLS)</i>			
WFH hours	0.212*** (0.040)	0.149** (0.061)	0.154** (0.061)
Women	-5.536*** (0.400)	-5.231*** (0.411)	-3.431*** (0.643)
College		0.930 (0.671)	0.994 (0.667)
Women × Kid			0.437 (0.957)
Women × Married			-3.639*** (0.815)
Education and experience controls	no	yes	yes
Family controls and interactions	no	no	yes
Observations	2627	2627	2627
Cragg-Donald Wald F	574.32	255.50	253.81

Notes: The dependent variable is desired weekly hours worked. Robust standard errors are reported in parentheses. * p<0.10, ** p<0.05, *** p<0.01. In Panel B, *WFH hours* is instrumented with *WFH feasibility*. All specifications include age, age squared, a gender indicator, and federal-state fixed effects.

TABLE C.5. Effect of WFH on Usual Hours Worked Full-Time

	(1)	(2)	(3)
<i>Panel A. OLS</i>			
WFH dummy	0.966*** (0.276)	0.252 (0.289)	0.247 (0.290)
Women	-0.999*** (0.250)	-0.985*** (0.251)	-0.841** (0.357)
College		0.891** (0.399)	0.957** (0.402)
Women × Kid			-1.266* (0.751)
Women × Married			-0.174 (0.517)
Education and experience controls	no	yes	yes
Family controls and interactions	no	no	yes
Observations	1765	1765	1765
R^2	0.035	0.056	0.060
<i>Panel B. IV (2SLS)</i>			
WFH dummy	2.767*** (0.596)	0.854 (0.896)	0.928 (0.895)
Women	-0.990*** (0.252)	-0.979*** (0.250)	-0.853** (0.354)
College		0.816** (0.409)	0.875** (0.411)
Women × Kid			-1.262* (0.750)
Women × Married			-0.150 (0.512)
Education and experience controls	no	yes	yes
Family controls and interactions	no	no	yes
Observations	1765	1765	1765
Cragg-Donald Wald F	511.07	199.94	199.97

Notes: The dependent variable is usual weekly hours worked. Robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. In Panel B, *WFH dummy* is instrumented with *WFH feasibility*. All specifications include age, age squared, a gender indicator, and federal-state fixed effects. The sample is restricted to observations with *full_month* = 1.

TABLE C.6. Effect of WFH on Desired Hours Worked Full-Time

	(1)	(2)	(3)
<i>Panel A. OLS</i>			
WFH dummy	0.759 (0.461)	-0.121 (0.505)	-0.170 (0.507)
Women	-2.511*** (0.466)	-2.567*** (0.470)	-2.408*** (0.725)
College		1.041 (0.748)	1.010 (0.752)
Women $\times$ Kid			2.394* (1.225)
Women $\times$ Married			-0.842 (0.940)
Education and experience controls	no	yes	yes
Family controls and interactions	no	no	yes
Observations	1765	1765	1765
R ²	0.038	0.049	0.051
<i>Panel B. IV (2SLS)</i>			
WFH dummy	2.475** (1.012)	-0.135 (1.692)	-0.188 (1.690)
Women	-2.503*** (0.464)	-2.567*** (0.467)	-2.407*** (0.721)
College		1.043 (0.771)	1.013 (0.773)
Women $\times$ Kid			2.394** (1.215)
Women $\times$ Married			-0.843 (0.935)
Education and experience controls	no	yes	yes
Family controls and interactions	no	no	yes
Observations	1765	1765	1765
Cragg-Donald Wald F	511.07	199.94	199.97

Notes: The dependent variable is desired weekly hours worked. Robust standard errors are reported in parentheses. * p<0.10, ** p<0.05, *** p<0.01. In Panel B, *WFH dummy* is instrumented with *WFH feasibility*. All specifications include age, age squared, a gender indicator, and federal-state fixed effects. The sample is restricted to observations with *full_month* = 1.

TABLE C.7. Effect of WFH Hours on UsualHours Worked Full-Time

	(1)	(2)	(3)
<i>Panel A. OLS</i>			
WFH hours	0.051*** (0.011)	0.031*** (0.012)	0.031*** (0.012)
Women	-0.984*** (0.249)	-0.972*** (0.250)	-0.838** (0.356)
College		0.831** (0.399)	0.898** (0.401)
Women × Kid			-1.307* (0.748)
Women × Married			-0.155 (0.514)
Education and experience controls	no	yes	yes
Family controls and interactions	no	no	yes
Observations	1765	1765	1765
R ²	0.043	0.060	0.065
<i>Panel B. IV (2SLS)</i>			
WFH hours	0.111*** (0.024)	0.033 (0.035)	0.036 (0.035)
Women	-0.962*** (0.249)	-0.971*** (0.249)	-0.838** (0.353)
College		0.824** (0.405)	0.884** (0.406)
Women × Kid			-1.313* (0.746)
Women × Married			-0.150 (0.509)
Education and experience controls	no	yes	yes
Family controls and interactions	no	no	yes
Observations	1765	1765	1765
Cragg-Donald Wald F	396.05	162.02	161.62

Notes: The dependent variable is usual weekly hours worked. Robust standard errors are reported in parentheses. * p<0.10, ** p<0.05, *** p<0.01. In Panel B, *WFH hours* is instrumented with *WFH feasibility*. All specifications include age, age squared, a gender indicator, and federal-state fixed effects. The sample is restricted to observations with *full_month* = 1.

TABLE C.8. Effect of WFH Hours on Desired Hours Worked Full-Time

	(1)	(2)	(3)
<i>Panel A. OLS</i>			
WFH hours	0.046*** (0.016)	0.022 (0.017)	0.020 (0.017)
Women	-2.498*** (0.465)	-2.555*** (0.470)	-2.411*** (0.725)
College		0.962 (0.747)	0.934 (0.751)
Women × Kid			2.370* (1.224)
Women × Married			-0.818 (0.940)
Education and experience controls	no	yes	yes
Family controls and interactions	no	no	yes
Observations	1765	1765	1765
R ²	0.041	0.049	0.052
<i>Panel B. IV (2SLS)</i>			
WFH hours	0.099** (0.041)	-0.005 (0.066)	-0.007 (0.066)
Women	-2.477*** (0.463)	-2.568*** (0.468)	-2.410*** (0.720)
College		1.042 (0.767)	1.011 (0.769)
Women × Kid			2.405** (1.216)
Women × Married			-0.843 (0.936)
Education and experience controls	no	yes	yes
Family controls and interactions	no	no	yes
Observations	1765	1765	1765
Cragg-Donald Wald F	396.05	162.02	161.62

Notes: The dependent variable is desired weekly hours worked. Robust standard errors are reported in parentheses. * p<0.10, ** p<0.05, *** p<0.01. In Panel B, *WFH hours* is instrumented with *WFH feasibility*. All specifications include age, age squared, a gender indicator, and federal-state fixed effects. The sample is restricted to observations with *full_month* = 1.

TABLE C.9. Effect of WFH on Usual Hours Worked Part-Time

	(1)	(2)	(3)
<i>Panel A. OLS</i>			
WFH dummy	3.104*** (0.734)	2.087*** (0.775)	2.053*** (0.759)
Women	-0.709 (1.068)	-0.082 (1.077)	0.510 (1.586)
College		1.368 (1.021)	1.685* (1.017)
Women × Kid			0.031 (2.636)
Women × Married			-0.801 (2.155)
Education and experience controls	no	yes	yes
Family controls and interactions	no	no	yes
Observations	825	825	825
R ²	0.115	0.145	0.172
<i>Panel B. IV (2SLS)</i>			
WFH dummy	9.974*** (1.642)	9.357*** (2.196)	10.282*** (2.210)
Women	-0.939 (1.113)	-0.584 (1.111)	-0.575 (1.585)
College		0.306 (1.083)	0.503 (1.097)
Women × Kid			-0.612 (2.763)
Women × Married			0.370 (2.246)
Education and experience controls	no	yes	yes
Family controls and interactions	no	no	yes
Observations	825	825	825
Cragg-Donald Wald F	218.32	120.60	119.65

Notes: The dependent variable is usual weekly hours worked. Robust standard errors are reported in parentheses. * p<0.10, ** p<0.05, *** p<0.01. In Panel B, *WFH dummy* is instrumented with *WFH feasibility*. All specifications include age, age squared, a gender indicator, and federal-state fixed effects. The sample is restricted to observations with *full_month* = 0.

TABLE C.10. Effect of WFH on Desired Hours Worked Part-Time

	(1)	(2)	(3)
<i>Panel A. OLS</i>			
WFH dummy	0.997 (0.719)	0.804 (0.752)	0.722 (0.741)
Women	-1.394 (1.007)	-1.117 (1.016)	1.017 (1.578)
College		-0.206 (1.034)	0.049 (1.033)
Women $\times$ Kid			-0.137 (2.298)
Women $\times$ Married			-3.628* (1.969)
Education and experience controls	no	yes	yes
Family controls and interactions	no	no	yes
Observations	825	825	825
R ²	0.091	0.097	0.125
<i>Panel B. IV (2SLS)</i>			
WFH dummy	3.234** (1.597)	3.730* (2.106)	4.508** (2.089)
Women	-1.469 (0.999)	-1.319 (1.001)	0.518 (1.542)
College		-0.634 (1.052)	-0.494 (1.058)
Women $\times$ Kid			-0.433 (2.327)
Women $\times$ Married			-3.089 (1.959)
Education and experience controls	no	yes	yes
Family controls and interactions	no	no	yes
Observations	825	825	825
Cragg-Donald Wald <i>F</i>	218.32	120.60	119.65

Notes: The dependent variable is desired weekly hours worked. Robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. In Panel B, *WFH dummy* is instrumented with *WFH feasibility*. All specifications include age, age squared, a gender indicator, and federal-state fixed effects. The sample is restricted to observations with *full_month* = 0.

TABLE C.11. Effect of WFH Hours on Usual Hours Worked Part-Time

	(1)	(2)	(3)
<i>Panel A. OLS</i>			
WFH hours	0.243*** (0.032)	0.201*** (0.033)	0.197*** (0.033)
Women	-0.706 (1.051)	-0.128 (1.061)	0.427 (1.576)
College		1.113 (0.999)	1.435 (0.992)
Women × Kid			-0.030 (2.599)
Women × Married			-0.721 (2.140)
Education and experience controls	no	yes	yes
Family controls and interactions	no	no	yes
Observations	825	825	825
R ²	0.136	0.163	0.189
<i>Panel B. IV (2SLS)</i>			
WFH hours	0.644*** (0.106)	0.574*** (0.133)	0.628*** (0.134)
Women	-0.871 (1.093)	-0.482 (1.081)	-0.347 (1.608)
College		0.075 (1.100)	0.239 (1.121)
Women × Kid			-0.513 (2.669)
Women × Married			0.098 (2.215)
Education and experience controls	no	yes	yes
Family controls and interactions	no	no	yes
Observations	825	825	825
Cragg-Donald Wald F	146.02	88.56	88.43

Notes: The dependent variable is usual weekly hours worked. Robust standard errors are reported in parentheses. * p<0.10, ** p<0.05, *** p<0.01. In Panel B, *WFH hours* is instrumented with *WFH feasibility*. All specifications include age, age squared, a gender indicator, and federal-state fixed effects. The sample is restricted to observations with *full_month* = 0.

TABLE C.12. Effect of WFH Hours on Desired Hours Worked Part-Time

	(1)	(2)	(3)
<i>Panel A. OLS</i>			
WFH hours	0.102*** (0.036)	0.096*** (0.037)	0.090** (0.036)
Women	-1.403 (1.000)	-1.152 (1.007)	0.951 (1.576)
College		-0.355 (1.022)	-0.096 (1.021)
Women $\times$ Kid			-0.181 (2.272)
Women $\times$ Married			-3.560* (1.961)
Education and experience controls	no	yes	yes
Family controls and interactions	no	no	yes
Observations	825	825	825
R^2	0.096	0.102	0.129
<i>Panel B. IV (2SLS)</i>			
WFH hours	0.209** (0.102)	0.229* (0.128)	0.275** (0.127)
Women	-1.447 (0.985)	-1.279 (0.985)	0.618 (1.546)
College		-0.726 (1.055)	-0.610 (1.064)
Women $\times$ Kid			-0.389 (2.243)
Women $\times$ Married			-3.209* (1.933)
Education and experience controls	no	yes	yes
Family controls and interactions	no	no	yes
Observations	825	825	825
Cragg-Donald Wald F	146.02	88.56	88.43

Notes: The dependent variable is desired weekly hours worked. Robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. In Panel B, *WFH hours* is instrumented with *WFH feasibility*. All specifications include age, age squared, a gender indicator, and federal-state fixed effects. The sample is restricted to observations with *full_month* = 0.

TABLE C.13. WFH and Usual Hours Worked, by Gender

	Men			Women		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. OLS</i>						
WFH dummy	1.631*** (0.453)	0.940* (0.516)	0.919* (0.515)	4.239*** (0.608)	2.698*** (0.626)	2.825*** (0.609)
College		1.245* (0.700)	1.305* (0.701)		2.738*** (0.875)	3.041*** (0.853)
Observations	1254	1254	1254	1373	1373	1373
R ²	0.0451	0.0565	0.0586	0.0741	0.1463	0.1983
<i>Panel B. IV (2SLS)</i>						
WFH dummy	4.039*** (0.978)	3.551** (1.787)	3.597** (1.817)	13.236*** (1.523)	10.119*** (1.966)	10.923*** (1.945)
College		0.913 (0.738)	0.993 (0.736)		1.640* (0.960)	1.856* (0.950)
Observations	1254	1254	1254	1373	1373	1373
Cragg-Donald Wald F	424.07	133.82	131.19	340.53	186.19	184.09

Notes: The dependent variable is usual weekly hours worked. Robust standard errors are reported in parentheses. * p<0.10, ** p<0.05, *** p<0.01. In Panel B, *WFH dummy* is instrumented with *WFH feasibility*. All specifications include age, age squared, and federal-state fixed effects.

TABLE C.14. WFH and Desired Hours Worked, by Gender

	Men			Women		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. OLS</i>						
WFH dummy	0.934 (0.583)	0.210 (0.668)	0.132 (0.668)	2.196*** (0.584)	1.355** (0.610)	1.416** (0.607)
College		1.161 (0.949)	1.158 (0.949)		1.253 (0.867)	1.428* (0.863)
Observations	1254	1254	1254	1373	1373	1373
R ²	0.0217	0.0279	0.0298	0.0651	0.0830	0.1038
<i>Panel B. IV (2SLS)</i>						
WFH dummy	1.952 (1.212)	-0.391 (2.295)	-0.478 (2.322)	7.952*** (1.351)	6.814*** (1.838)	7.248*** (1.829)
College		1.238 (0.997)	1.229 (0.991)		0.445 (0.921)	0.575 (0.921)
Observations	1254	1254	1254	1373	1373	1373
Cragg-Donald Wald F	424.07	133.82	131.19	340.53	186.19	184.09

Notes: The dependent variable is desired weekly hours worked. Robust standard errors are reported in parentheses. * p<0.10, ** p<0.05, *** p<0.01. In Panel B, *WFH dummy* is instrumented with *WFH feasibility*. All specifications include age, age squared, and federal-state fixed effects.

TABLE C.15. WFH Hours and Usual Hours Worked, by Gender

	Men			Women		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. OLS</i>						
WFH hours	0.081*** (0.014)	0.064*** (0.016)	0.063*** (0.016)	0.258*** (0.021)	0.204*** (0.022)	0.204*** (0.022)
College		1.130 (0.695)	1.192* (0.696)		2.552*** (0.860)	2.871*** (0.834)
Observations	1254	1254	1254	1373	1373	1373
R ²	0.0527	0.0626	0.0645	0.1086	0.1729	0.2238
<i>Panel B. IV (2SLS)</i>						
WFH hours	0.165*** (0.040)	0.139** (0.070)	0.141** (0.070)	0.646*** (0.075)	0.493*** (0.096)	0.530*** (0.095)
College		0.850 (0.744)	0.917 (0.743)		1.725* (0.941)	1.939** (0.935)
Observations	1254	1254	1254	1373	1373	1373
Cragg-Donald Wald F	327.99	110.03	108.25	234.70	128.18	127.62

Notes: The dependent variable is usual weekly hours worked. Robust standard errors are reported in parentheses. * p<0.10, ** p<0.05, *** p<0.01. In Panel B, *WFH hours* is instrumented with *WFH feasibility*. All specifications include age, age squared, and federal-state fixed effects.

TABLE C.16. WFH Hours and Desired Hours Worked, by Gender

	Men			Women		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. OLS</i>						
WFH hours	0.061*** (0.019)	0.043** (0.021)	0.041* (0.021)	0.141*** (0.024)	0.113*** (0.024)	0.112*** (0.024)
College		1.029 (0.944)	1.030 (0.944)		1.130 (0.859)	1.314 (0.855)
Observations	1254	1254	1254	1373	1373	1373
R ²	0.0256	0.0303	0.0320	0.0778	0.0928	0.1132
<i>Panel B. IV (2SLS)</i>						
WFH hours	0.080 (0.049)	-0.015 (0.090)	-0.019 (0.091)	0.388*** (0.066)	0.332*** (0.090)	0.352*** (0.089)
College		1.245 (1.012)	1.239 (1.008)		0.502 (0.912)	0.630 (0.912)
Observations	1254	1254	1254	1373	1373	1373
Cragg-Donald Wald F	327.99	110.03	108.25	234.70	128.18	127.62

Notes: The dependent variable is desired weekly hours worked. Robust standard errors are reported in parentheses. * p<0.10, ** p<0.05, *** p<0.01. In Panel B, *WFH hours* is instrumented with *WFH feasibility*. All specifications include age, age squared, and federal-state fixed effects.

TABLE C.17. Placebo robustness: extensive-margin treatment only

	Actual hours	Desired hours	Hours gap
<i>Panel A. Bottom 10% of WFH feasibility</i>			
OLS: WFH dummy	2.126 (1.902)	3.747 (2.731)	1.357 (3.400)
IV: WFH dummy	121.852 (621.236)	586.287 (1975.655)	348.167 (1235.011)
Cragg–Donald Wald F	0.04	0.04	0.04
Anderson–Rubin p -value	0.807	0.271	0.507
Observations	285	285	285
<i>Panel B. Bottom 25% of WFH feasibility</i>			
OLS: WFH dummy	2.498 (1.988)	3.854** (1.805)	1.338 (1.552)
IV: WFH dummy	-67.613 (185.954)	-70.469 (185.669)	-28.107 (155.325)
Cragg–Donald Wald F	0.20	0.20	0.20
Anderson–Rubin p -value	0.662	0.645	0.855
Observations	665	665	665

Notes: Each column reports the treatment coefficient only. OLS rows are from robust OLS regressions. IV rows are from 2SLS using `wfh_feas` as instrument. All placebo specifications use control set (3) only. The placebo samples are restricted to the bottom 10% and bottom 25% of the WFH-feasibility distribution. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE C.18. Placebo robustness: intensive-margin treatment only

	Actual hours	Desired hours	Hours gap
<i>Panel A. Bottom 10% of WFH feasibility</i>			
OLS: WFH hours	0.022 (0.086)	0.185 (0.125)	0.145 (0.130)
IV: WFH hours	1.561 (6.236)	7.511 (8.769)	4.460 (7.186)
Cragg–Donald Wald F	0.52	0.52	0.52
Anderson–Rubin p -value	0.807	0.271	0.507
Observations	285	285	285
<i>Panel B. Bottom 25% of WFH feasibility</i>			
OLS: WFH hours	0.163** (0.075)	0.235*** (0.069)	0.071 (0.084)
IV: WFH hours	-3.950 (10.435)	-4.117 (10.463)	-1.642 (9.037)
Cragg–Donald Wald F	0.18	0.18	0.18
Anderson–Rubin p -value	0.662	0.645	0.855
Observations	665	665	665

Notes: Each column reports the treatment coefficient only. OLS rows are from robust OLS regressions. IV rows are from 2SLS using `wfh_feas` as instrument. All placebo specifications use control set (3) only. The placebo samples are restricted to the bottom 10% and bottom 25% of the WFH-feasibility distribution. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE C.19. Rich-covariates robustness: Usual Hours Worked

	Full sample		Women	
	Extensive margin	Intensive margin	Extensive margin	Intensive margin
IV (2SLS)	7.282	0.315	10.923	0.530
IVR	7.309	0.318	11.009	0.536
Observations	2627	2627	1373	1373

Notes: The dependent variable is usual weekly hours worked. Each entry reports the treatment coefficient only. In the extensive-margin specifications, the treatment is the WFH dummy. In the intensive-margin specifications, the treatment is WFH hours. All columns report the IVR estimate, where the instrument is the residualized instrument and the specification includes control set (3).

TABLE C.20. Rich-covariates robustness: Desired Hours Worked

	Full sample		Women	
	Extensive margin	Intensive margin	Extensive margin	Intensive margin
IV (2SLS)	3.552	0.154	7.248	0.352
IVR	3.612	0.157	7.330	0.357
Observations	2627	2627	1373	1373

Notes: The dependent variable is desired weekly hours worked. Each entry reports the treatment coefficient only. In the extensive-margin specifications, the treatment is the WFH dummy. In the intensive-margin specifications, the treatment is WFH hours. All columns report the IVR estimate, where the instrument is the residualized instrument and the specification includes control set (3).

TABLE C.21. Unrestricted-sample robustness: Usual Hours Worked

Sample	Margin	OLS	2SLS	Observations	Cragg-Donald Wald F
<i>Panel A. Full sample</i>					
	Extensive margin	1.894*** (0.350)	7.285*** (1.160)	4347	514.89
	Intensive margin	0.143*** (0.012)	0.324*** (0.051)	4347	389.77
<i>Panel B. Part-time</i>					
	Extensive margin	1.505** (0.601)	8.603*** (1.976)	1518	170.07
	Intensive margin	0.218*** (0.026)	0.555*** (0.125)	1518	128.09
<i>Panel C. Women</i>					
	Extensive margin	2.620*** (0.519)	11.677*** (1.789)	2379	253.40
	Intensive margin	0.220*** (0.018)	0.595*** (0.091)	2379	172.81
<i>Panel D. Men</i>					
	Extensive margin	1.201*** (0.450)	3.299** (1.499)	1967	241.76
	Intensive margin	0.029*** (0.010)	0.032 (0.029)	1967	192.59

Notes: The dependent variable is usual weekly hours worked. All columns report estimates from specification C3 only. In the extensive-margin columns, the treatment is the WFH dummy. In the intensive-margin columns, the treatment is WFH hours. OLS rows report robust OLS estimates and 2SLS rows report IV estimates using `wfh_feas` as instrument. The part-time sample is defined by `full_month = 0`. Robust standard errors are reported in parentheses. Cragg-Donald Wald F statistics are reported for the 2SLS specifications. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE C.22. Unrestricted-sample robustness: Desired Hours Worked

Sample	Margin	OLS	2SLS	Observations	Cragg–Donald Wald F
<i>Panel A. Full sample</i>					
	Extensive margin	0.887** (0.352)	3.607*** (1.155)	4347	514.89
	Intensive margin	0.080*** (0.012)	0.160*** (0.051)	4347	389.77
<i>Panel B. Part-time</i>					
	Extensive margin	0.859 (0.566)	4.026** (1.856)	1518	170.07
	Intensive margin	0.120*** (0.028)	0.260** (0.119)	1518	128.09
<i>Panel C. Women</i>					
	Extensive margin	1.500*** (0.481)	7.557*** (1.638)	2379	253.40
	Intensive margin	0.124*** (0.019)	0.385*** (0.084)	2379	172.81
<i>Panel D. Men</i>					
	Extensive margin	0.200 (0.514)	-0.282 (1.696)	1967	241.76
	Intensive margin	0.023** (0.009)	0.012 (0.027)	1967	192.59

Notes: The dependent variable is desired weekly hours worked. All columns report estimates from specification C3 only. In the extensive-margin columns, the treatment is the WFH dummy. In the intensive-margin columns, the treatment is WFH hours. OLS rows report robust OLS estimates and 2SLS rows report IV estimates using `wfh_feas` as instrument. The part-time sample is defined by `full_month = 0`. Robust standard errors are reported in parentheses. Cragg–Donald Wald F statistics are reported for the 2SLS specifications. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.